

Lec 60 Bidirectional Encoder Representations from Transformer(BERT)

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Transcript

[00:00] Hello everyone, welcome back to the lecture series on fundamentals of generative AI and large language models. In my previous session, we have discussed about the transformer architecture, the encoder and the decoder part. In today's session, we will talk about bidirectional encoder representations from transformer which we call as BERT. Okay, BERT is a language model. So we will see what are the contents for today's discussion. The contents goes like motivation for BERT, introduction to BERT, why does BERT use only encoder, pre-training and fine-tuning of language models, BERT pre-training tasks, most language modeling that is the first one in B pre-training tasks and next sentence prediction. Then we will have a look

[00:55] into BERT pre-training and five fine-tuning pipeline and finally the variance of BERT. Okay. So we will start with what is the motivation behind the development of BERT. Okay. So before birth whatever language models that people were using like you know statistical language models like n-gram language model there are certain limitations of those language models. What are they? Suppose just consider this

example. I deposited money in the bank. So this is one sentence. I deposited money in the bank which means this bank basically is a financial institution whereas the children played near the bank of the river it means this bank belongs to river bank. So the meanings are very different but in earlier

[01:55] language models there is a static word embedding using word toe or glove embedding. There is no difference in the embedding that is there for this bank which is the financial institution and the bank that is that means the river bank. Okay. So that becomes a limitation. So there is no contextual meaning that is there with the word with the context with what context this word is coming that was missing. So it is a static word embedding. So this was one of the limitations of the earlier language models.

[02:34] Secondly, all these earlier language models, they were reading the text from left to right. So they were processing left to right each and every token. Okay. The movie was dash. Okay. Now as they are moving from left to right they see this movie was dash the movie was good the movie was bad both is possible okay now without knowing this one because they are reading it or processing it from left to right what happens is they ignore the information that appears later in the sentence so we don't know what the sentence's sentiment is the movie is good which gives a positive sentiment. The movie is bad it gives a negative sentiment. Okay. So as we were processing from left to right the information which is coming or

[03:36] appearing later in the sentence they were missing. Okay. So hence this is a limitation in understanding what this sentence sentiment is. So this is one one of the limitations poor long range understanding. So previously RNN's and LSTMs we have seen they often struggle to connect words that are far apart in the long sequence. So if there is a relationship between the second word and the 10th word there may be they may miss or the they struggled to connect these two words.

[04:14] That means the long range dependency to some extent it was captured by LSTM but long long range dependency was a problem still a problem. Then the fourth or one of the very important uh limitations was each and every task required a separate model to be trained. when I tell each and every separate uh task it may be sentiment analysis, it can be question answering, it can be uh you know uh neer name entity recognition. So each of these models if I want the model to do a sentiment analysis I need a separate model. If I need to do question answering I need a separate model. If I need to do named entity recognition I need a separate model. So task specific models were required.

[05:08] So these are the main challenges which motivated scientists to think of something where we can overcome these limitations. Okay. And that is where our birec-tional encoder representation from transformer or the bird came into picture. So what is this birth? Birectional encoder representation from transformers is basically

a pre-trained deep learning language model developed by Google in 2018. Okay. So this is the reference for the first paper on BERT.

[05:51] Okay. Develin and his team. The name of the paper is pre-training of deep bidirectional transformers for language understanding. Okay. So this came into picture. It got published in 2019 but came into existence in 2018. Okay. And we will see there are two terms that I have marked in red. One is the pre-trained and other one is the language model. So we will look into what are these pre-trained and what is this language model. Okay. So this architecture B architecture is entirely built on the trans transformer encoder architecture. So we have seen in our last sessions encoder and decoder these are the two parts or building blocks of a transformer.

[06:42] Now this bird is or bird uses only the encoder part of the transformer not the decoder. Okay. and it is designed to understand words considering both left and right context simultaneously. So if the sentence is the cat sat on the mat, the cat sat on the mat. The model will read it left to right as well as from right to left so that the model gets the context what is there at the end also and what is you know coming from the beginning also from left to right as well as from right to left. Okay, this gives a much better contextual understanding of the sentence.

[07:34] Now what is this language model? Okay, and what do I mean by pre-trained? we will come there. So basically what is a language model? Because this language model we will be keep on telling telling telling from now onwards we'll keep on you know using it. Language models large language models. Okay. So let us have a quick recap on what exactly this language model is supposed to do or what is our expectation from the language model. So a language model learns the probability distribution over a sequence of words. That means a language model what it will do is if there is a sequence of words okay then it will give some probability value to a specific to all the words that that may come in that sequence and the one which is having

[08:28] highest probability will come as the next word. Okay. So language models learn the probability distribution. You have a vocabulary. You have some part of a sentence and then there is there is in that sentence there is some sequence of words that is left to be filled and your model what is it doing? It is giving some probability score to all the words that are present in the vocabulary and whichever word is getting the highest score that comes as the next sentence.

[08:57] That means there is some logical flow in the sentence. Okay. So that is what the language models do. Now traditional language models perform next word prediction. As I was telling there is a sequence of words. Half of the words is given as the input sequence. Another one or two are there or next one what will be the best word that will be predicted by your language model. Okay. And how the language model learns? It learns by analyzing billions of text examples. it analyzes or it gets

trained using billions of text documents or examples and then it learns the statistical pattern and relationship between the words. So there is a language understanding which happens inside the model and due and because of

[09:46] that training it is able to predict the next word. This is in the case of traditional language models. Okay. Now the question comes in bird why do we use only the encoder why not the decoder? Okay why not the encoder decoder together. So the primary goal of bird is to understand the language not to generate language. Okay. So let us have a look there what exactly we mean when we tell understand language and not generate language.

[10:22] So here I have listed couple of applications of bird like question answering, sentiment analysis, named entity recognition, uh text classification. So these are some applications of bird. Now any of these tasks or all of these tasks they do not require writing a sentence. Okay, we will see this one more in details in my next slide when we go with examples. It will become much more clear at that time. They only require understanding the existing text.

[10:57] Suppose you want to do sentiment analysis. You what what output do you want your model to give? It will tell you either it is positive or negative or neutral. So there is no sentence generation that is happening. Okay? There is no language generation. There is just a small word that is coming as the output. The same is the case for all the other applications. We will have a look to each of them now. Now in case of sentiment analysis, suppose my input word is I absolutely love this movie.

[11:29] What do you want your model to give you? This is the model's output. It is positive sentiment. That means you expect only one word here, not a full sentence. Okay. Come to named entity recognition. If the input is Dr. Joan visited Ames Hospital on August 1st, 2025. And what what output do you expect your model to give? Like the entity is Dr. Joan is a person, Ames Hospital is an organization, 1st August 2025 is a date.

[12:01] So here also you see there is no generation of language. It is just the answers to the questions. Okay. Question answering. Suppose the input is a paragraph. Paris is the capital of France and what is the question that is given? What is the capital of France? So in question answering what we do is we give a paragraph to the model and then we pose a question to the model. Based on the paragraph the understanding of the paragraph. The model now will give the answer to the specific question. So in this case the question is what is the capital of France? And the output that I expect the model to give me is Paris.

[12:41] Okay. So here in all these examples we have seen there is what is required from bird is the understanding of the text that is given to it not the generation. Okay because in the output we are expecting one word one or two words there is no complete generation of language there that is the reason B uses only the encoder. Okay.

Now we will talk about two terms pre-training and fine-tuning of a language model. Okay. So what do I mean by these these two words in bird we have told that it is a pre-trained language model. So many modern pre-trained language models like bird GPT they are trained in two stages. There is two stage training. What is this? First one is called as the pre-training where the the model learns the

[13:43] language, understands the grammar, understands relationship between words and the context of the language and it is learned from a massive corpus of text. So you give all documents all every text document that is present you know with you that you have access to and you give it for training the model. Okay. So it is basically we can uh you know think in in think it in this direction like when a kid is studying in school from class KG to class 10 or 12 they are trained in a lot of subjects.

[14:30] Okay. So through all these trainings they learn languages, they learn mathematics, they learn science, they learned history, geography, arts, everything. Okay. So through this process the child learns or understands in general a generalized knowledge base is created and when they go for higher studies if anyone is going for medicals you know medical school they will be trained in a specific you know a stream they will be trained with what are the diseases how to diag diagnose them, how to treat them, what are the symptoms. So it's a specialized track. So pre-training we can consider is it as your class KG to class 12 you know uh studies whereas fine-tuning is when you are fine-tuning your model for a

[15:31] specific task as I was telling when you go and study medical you are specialized to become a doctor. Okay. So that's a specialized uh uh stream of study. So pre-training and fine-tuning exactly in the same way you can relate them. So training a language model with lot of uh information about the grammar, the context, the vocabulary of a language. So in this case what they learn is the language understanding. fine-tuning takes a pre-trained language model and trains it further on a specific task. As I was telling anyone who wants to go for medical studies, they will take that student or the student who have the knowledge base from class KG to class 12 then they will go for the specialized track of medicine.

[16:24] Okay. And here also in fine-tuning the pre-trained language model is taken and then it is trained on a specific task. When you are training it for a specific task you are training it in a smaller labeled data set. Okay. And then it becomes an expert in the task. So when you are you know um suppose you want a model which will be very specialized in document classification like whether it is a sports news whether it is an entertainment news whether it is an educational news the your model will be able to tell that one. So you take a pre-trained model which have a general understanding of the language of the context uh in general the grammar and everything and then you take labeled data and train it specifically for that text

[17:20] classification. Okay. So that is the concept of fine-tuning. So bird is a pre-

trained language model. Okay. So we now know what do I mean by pre-trained. We know what is language model what what do we expect from a language model to do what does a language model do okay and we have understood what is pre-training and fine-tuning now when bird is getting pre-trained there are some specific tasks through which bird gets pre-trained so this pre-training task is basically a self-supervised learning there is no supervision that is there.

[18:02] It is a self-supervised learning with the objective. What is the objective of this self-supervised pre-training? There are two specific objectives for which it is trained and it is trained from a large amount of unlabeled text. Okay. So we will see what are those two pre-training tasks. The first one is masked language modeling. We call it as MLM, [clears throat] masked language modeling. What exactly it is? You know, we mask some words in a sentence and then we train the model to predict that masked word. Okay, we will look into it in details in my next slide. So here we will just have a look into two important points. Traditional language models predict the next word. As we have already seen the cat sat on the dash

[18:57] that means it will predict the cat sat on the mat. Okay, which limits them often from left to right context understanding. So as I was telling traditional uh language models they always process the words from left to right and in this process what happens is you may miss you know what is there miss the information that is coming after that after that word. Okay. So bird instead learns by predicting randomly masked words. So if you have a sentence, we will mask or we will cover up some particular words and then we will tell bird to use both left and right context. It will read it both from left side and right side and then what it will do? It will find out what is the it will predict the masked word.

[19:49] Okay. The second task that is there is next sentence prediction. Okay, what is that? We will have a look. This is having application in question answering, natural language inference, document retrieval. We will see what exactly it is. Okay. So this next sentence prediction requires understanding the relationship between two sentences. Suppose I have sentence A and I have another bunch of sentence B, C, D. I the model will check which of these sentences is logically fit to come and sit as the sentence after sentence A. So it requires understanding the relationship between two sentences.

[20:38] Okay. So this next sentence prediction trains bird to determine whether a sentence is logically following the other. So that is about next sentence prediction. So we will look into what is this masked language modeling. Suppose we have the original sentence as the boy is playing football. This is my original sentence. Now this the input to the birth I'm giving from the original sentence is the boy is this word is masked football boy is dash football okay I have masked that word and I want bird to read it from left side as well as from right side and predict the m word okay so the target output is playing the boy is playing football. That is the target output. Okay.

Now bird predicts the masked word using information from

[21:40] both sides. That is where bidirectional thing comes into picture. Okay. Now what is the masking strategy here? When I'm telling I'll mask some words and then train the model to predict those masked words. What is the masking strategy? So in the original paper they have used the strategy like 15% of the tokens are randomly selected. Of these selected tokens 80% are replaced with the word with mask token. Okay. So you have say 100 tokens out of that 15 tokens you select and from that 15 80% of that 15 is replaced with a mask token.

[22:28] 10% is replaced with a random word. 10% of that 15 I am just putting random words. I'm not masking them. And another 10 10% is left unchanged. So this is the masking strategy. Now why this masking strategy? Why we are replacing in some places the mask with the random word token? Because during the inference time during the you know uh when after the model is trained during the inference time what happens is the mask token does not appear so if I train only with mask tokens the model will get confused during inference so 10% of that we have replaced with random words and then we have trained so that there is no mismatch between pre-training and fine-tuning.

[23:25] Okay. And they get they the model is able to predict the masked words properly. Okay. So basically what the the idea here is you mask certain words and from the and some of the words instead of masking it you put some random other words. So the model now knows okay it is not the exact sequence everywhere correct sequence everywhere some random words are also there as filler and there are some some places where it is masked.

[23:59] So there is a much better training that happens to the model in predicting the masked words. Okay. Now bird learns contextual word representation from this masked language modeling. Okay, then we will come to this next sentence prediction. What exactly it is? Suppose I have sentence A. Rahul went to the grocery store and there is sentence B. He brought he bought some milk. Okay, these are the two sentences. What does next sentence prediction predict? It will give a level is next.

[24:39] That means sentence B is or can follow sentence A or sentence B follows sentence A because there is a logical relationship between A and B. Okay. So the label that is predicted by next sentence prediction is is is next. Think of another situation. Rahul went to the grocery store and Eiffel Tower is in Paris. So there is no matching. There is no logical you know relationship between these two words. So the label that is coming out is not next. Okay.

[25:13] So from next sentence prediction B learns the sentence level relationship because lot because we give paragraphs to the model to understand. So in a paragraph there will be multiple sentences coming one after the other. So the the model learns whether sentence B can follow sentence A or whether it is out of any relationship. If

it is any if if there is no relationship then the label come coming out is not next. And during training process particularly during pre-training 50% of the sentence pairs are actually consecutive sentence is next 50% are not next. Okay. And B learns to predict whether sentence B follows sentence A as I have told you explained you. Okay. So this is about next sentence prediction.

[26:08] Then we will look into bird training pre-training pipeline. What exactly happens in the pre-training pipeline? So we have the training data which is books corpus plus Wikipedia. It's a you know big corpus and then these training data under goes the subword tokenization like word piece tokenization and then there is an input construction we'll look into it little bit more in details in this slide as well as in the next slide and after that whatever we give input to the bird it goes through two pre-training task the mass language modeling plus the next sentence prediction Okay, both are there. And then finally we get the pre-trained birth. Okay, so you have uh book corpus, a lot of books, Wikipedia, any other re relevant

[27:06] documents and then it is undergoing subword tokenization using wordpiece tokenization. Then you get the input embeddings and then those input goes into the bird model. Then mass language modeling and sentence prediction is learned as the pre-training tasks or objectives. And then we get the pre-trained bird model. Okay. Now we will look into this input construction. What do we mean by that? So your input construction is basically it's an addition of three different types of tokens. One is the normal token embeddings. Okay, vector embeddings.

[27:44] Then there is something called as a segment embedding and position embedding. So what are these? We will have a look now. So the token embeddings these are basically the word piece token embeddings. Then we have the position embeddings which is the which encodes the words order. We have seen during uh you know learning the transformer architecture. What is this positional encoding or positional embedding does? It encodes each in which order the words are coming. Okay. And together with that one we have this segment embedding which distinguishes sentence A and sentence B.

[28:23] So there are two sentences. This segment embedding will tell which part is coming from which sentence. Okay. So your input embedding is a sum of the token plus position plus segment. All these embeddings together. Okay. So this is a picture I have taken from the original paper. So if this is the input to the bird input to bird B birth model you see there is position embeddings $E_0 E_1 E_2 E_3 E_4 \dots E_{10}$. So this is these are telling the position of the words in the sent in the text. Okay. Then added to that one these are segment embeddings. If you see EA EA EA EA EA EA then EB E E that means there are two sentences sentence A and sentence B there are two sentences there. Okay. So the embeddings that are coming from

[29:27] sentence A they have written as EA and embeddings that are coming from

sentence B have been written as EB. Okay. Then there are token embeddings which is basically your word piece word piece token embeddings. So [snorts] this is the embedding for my dog embedding for ease embedding for cute embedding. This is a separator. There are two new tokens you will see here. One is e cls classification token. It's a spe special kind of token which is usually given to the uh the starting of any input sequence here and this is the separator token which separates out between two sentences. So my dog is cute. So there is a separator and this is another sentence he likes playing playing. Then there is a separator uh token once more.

[30:22] So these are the special tokens that are used uh while putting the input to the bird. Okay. And we will see I have taken this picture from the original paper the reference is here. Okay. And in but so this is about the uh you know the pre-training and to in pre-training what is happening or how the input representations are created for for giving input to the bird model. Then we will look into bird fine-tuning pipeline. So in fine-tuning we all know it is a process of adapting a pre-trained bird to a specific NLP task.

[31:01] The NLP task can be um sentiment analysis, it can be question answering. Okay. So it can be um named entity recognition anything. Okay. So you have a pre-trained uh BERT model which has been you which has been pre-trained with the mask uh language modeling and um uh with the next sentence prediction. Okay. Um so with these two they have been trained and then you have a pre-trained B model and then you want it to be fine-tuned specifically for say question answering. So what do you do? A small task specific output layer is added.

[31:43] Okay. So we will see in the next picture how this output layer maybe a you know linear layer is added at the top of the entire architecture. This is very task specific. Whether it is if you want question answering you have to put a layer there which is relevant to question answering. If you want sentiment analysis, that kind of thing should go in the layer added layer after the pre-trained bark architecture layer uh burk architecture. Okay. And the entire burke model and the newly new layers are trained together on labeled data set. Two things I want you all to you know um have a focus on. One is this one is the label data set which means it is a supervised learning.

[32:28] Okay, it is supervised learning here. and the new layers and the bird model and the new layers are trained together. Now the question that you may ask me is if I am training my bird model entirely once again why do I need the pre-training? The answer is like this. We are or the bird pre-trained models weights are updated with the specific label data set training. So it is not training the bird model from the scratch but updating certain weight weights of the bird model so that it becomes very specific for that particular task. And this training the finetuning is done with label data set. So we have seen before that our pre-training was basically self-supervised whereas our fine-tuning is supervised with labeled

[33:29] data. Okay. So the flow is like this. We have the pre-trained bird model then add task specific output layers and then train on a small labeled data set and you get a task specific model. Okay. So what is the takeaway here? You have pre-trained the B birth B model and from that same model you are just adding some you know extra layers for specific tasks. That means your model is same only the last output layers or last layers of the entire architecture is changed to cater to different type of specific tasks of NLP. Okay, pre-training is self-supervised while fine-tuning is supervised. This I have already told now I have taken this picture once again from the paper original paper. This is regarding pre-training and this is

[34:30] fine-tuning. So you see here it is let us see at this picture current uh uh you know first. So you have unlabelled sentence A and B pair unlabelled sentence because we don't want supervised learning or we don't need bird doesn't need supervised learning in pre-training process. So we have this unlabelled data and then we have this masked sentence A and masked sentence B that means certain words of sentence A and sentence B are masked and that is given as input to your B model. So these are basically token you know tokens that are given the pink boxes. So this is the cls token and this is the separator token from these are from sentence A and these are all the tokens from sentence B. So token one to token M. Okay. So

[35:26] this is the tokens going from two sentences. Okay. And then these are the embeddings. Okay. and which are going to e cls e1 e2 e n to e1 e e1 dash to em dash with a separator embedding also. So all these things are going inside your birth model. So all these things are going inside and then these are basically your predictions from the birth. What is this next sentence prediction? So this C is basically your CLS and then T1 to TN then separator T T1 to TM prime these are the masked language models. Okay. So these are prediction of the masked language model that means it is predicting the masked words which are given to the input. Okay. So this is the pre-training part. So you have tokens they are going for embedding and then uh

[36:25] they are you know uh giving you the predictions for two two specific targets are here objectives of pre-training are there one is the next sentence prediction another is the mass language prediction okay m language model prediction that is the prediction of the masked words now we will look into the finetuning so finetuning here it is shown for It is uh for question answering. Okay. And you have given a question here. Question answer pair is given as the input to the model. The question is given here. The paragraph which is having the context is given here. Once again the cls token here is the start which shows the start of the token start of the sequence and then token one to token n here token one to

[37:14] token m it is given. Then you are giving these embeddings to bird and what is the output that you are you know uh getting or the predictions from here is the start and end span of the you know where exactly your um answer to the question will be there. Okay. So that is the prediction and this is done for this is the question

answering uh data set and then it can be done for these are very task specific. So this is question answering this is named entity recognition this is uh natural language inference multigenra natural language inference. So these are various specific tasks for which the fine-tuning can be done. Now there are some dotted arrows here. What exactly it means? That means the pre-trained weights are taken here

[38:10] from the pre-trained model. These pre-trained weights are going to the fine-tuning for for fine-tuning and there because uh task specific tuning will be done in the finetuning process. Okay. So this is about pre about pre-training and finetuning and the cls token is a special token added at the beginning of the input to the bird which I have told whose final embedding serve as a representation of the entire sequence for the classification task. Okay. So now there are couple of this this is the fundamental of BERT architecture. Now there are a couple of you know variants of BERT like Roberta which is the robustly optimized BERT approach. So it removes next sentence prediction and trains on more data with larger batches

[38:56] and dynamic masking. So this is one variant Alberta is there a light B basically it reduces the model size and memory usage through parameter sharing. Okay, so that is done in Alberta due to which the model size reduces keeping the performance high. Then we have distilled bird. It is a compressed version of B bird B that is faster and smaller retaining about 95% of bird's performance with fewer parameters. So it's a compressed version. And then we have Electra which is the efficiently learning and encoder that classifies token replacement accurately. This is another variant of bird. It trains bird more effectively by detecting the replace tokens instead of predicting mask tokens. So it will not predict the

[39:46] mask tokens but detect the replaced tokens and it will achieve stronger performance with lesser computation. So these are the variants of BERT but uh and with this one we have understood how the BERT architecture works and what are the advantages of BERT. Okay. So with this I I come to the end of today's session before I conclude just a quick recap. So we started with what are the limitations of u our traditional language models left to right context there was no right to left context then birectional thing was missing and uh then we had uh you know problem in with the static word embeddings that means a similar word can have different meanings from different context when they are put in different context then long-term

[40:40] dependencies as well as task task specific uh models was not there. Each and every task required a specific model. So from there bird came into picture which uses the encoder part of the the transformer using birectional you know encoding and birectional context learning and it is trained through two processes. It is a pre-trained model. So the training stages are pre-training and fine-tuning for any language model. And in pre-training a generalized u uh training of the language is given whereas fine-tuning is basically for specified tasks. So we have seen what is the pipeline for bird for pre-training what are the pre-training objectives of bird.

[41:26] So all these things we have seen masked language modeling and the next sentence prediction. These are the two pre-training tasks that Bert is supposed to learn during pre-training and then it is fine-tuned for specific tasks like question answering, named entity recognition, sentiment analysis, so on and so forth. So with this one, I come to the conclusion of today's session and in my next session we will be talking about the generative pre-trained transformers which is the which is the GPT. So uh we will take you from BERT to GPT or what is the you know differences between BERT and GPT and uh with that one um I will go forward with GPT's architecture and other things in the sess next session. So till we meet with

[42:12] the GPT architecture and GPT's learnings uh I urge all of you all the participants to go through the original paper of bird because you will get lot more information lot more details about the architecture and other things that they have used in that paper. Please go through it and till we meet for the next session on GPT happy learning and thank you all.